AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 18 · STUDENT LESSON

Untrusted input boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Validate the whole input. Check trusted scope. Look up data, never execute it.

Apply exact ASCII input contracts while keeping format, authority, record existence and execution boundaries separate.

Keep one habit: Input/local-data contract → four gate paths → lowercase picture → full-match/type probes → executable dictionary audit → independent uppercase set.

MAIN LESSON · SECTION 1

Keep input data from becoming instructions or permission

> An accepted shape is only one gate on the way to a supported answer.

Learning goals

I can explain the **mechanism** and terminology of **untrusted input** boundaries.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve trusted evidence and recovery boundaries

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall External side effects and recovery. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

Explain **untrusted input** boundaries using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **untrusted input** boundaries

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Declare the exact input contract and the separate authority source

> The added dictionary fixture makes the investigation complete without implying a real room record.

Begin with data, shape and access as different concepts

A room identifier is data. Even if a supplied string looks like a command, the lookup service must not run it as code. This exercise accepts exactly two ASCII characters: one uppercase letter and one digit. It does not trim spaces, change case or convert other types. After shape validation, the service still checks trusted access and looks up the exact key. A correct-looking identifier cannot authorize itself.

Define untrusted input boundaries

Untrusted input must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

Complete main contract and local lookup givens

Main uppercase contract and local extension	Complete rule or fixture
Input type and shape	Exact Python str, whole input matching [A-Z][0-9]
Normalization	None: no trimming, case conversion or coercion
Trusted allowed IDs in the local extension	A1 and D4, supplied by the test harness

Main uppercase contract and local extension	Complete rule or fixture
Local records	A1 has capacity 12/source ROOM-A1; B2 exists but is not in the allowed scope
Authorized but absent ID	D4 returns review after a permitted lookup
Response fields	Only status, room_id, capacity and source_id; record fields are checked
Instruction-shaped input	delete_all() fails shape; no input is evaluated as code
Observation	Lookup-call IDs and an unchanged source snapshot
Scope	Fictional records and trusted-scope fixtures, not real authentication or universal injection protection

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Trace each gate without treating a string as control

> A format pass, an authorization pass and an existing record establish different facts.

Original exact-room-ID and dictionary-lookup mechanism

For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Complete four-path local decision trace

Validate shape, then authorize, then look up data

SUPPLIED INPUT	CHECKS AND PERMITTED LOOKUP	LOCAL RESULT
A1	Whole shape passes; trusted scope allows; exact record exists	Answer capacity12 with source ROOM-A1
delete_all()	Whole shape fails before access or lookup	invalid_input; no instruction executes
B2	Shape passes, trusted access excludes it	denied; no record lookup
D4	Shape passes and access allows, but record absent	review; do not invent a capacity

The original worked case classifies A1 and instruction-shaped text. This explicitly supplied local extension additionally tests authority, existence and response fields.

State the contract before checking

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Untrusted input boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change case or length while keeping the contract fixed

> Do not silently normalize a failing value to make it pass.

Current task and response

- **Input**s A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed lowercase-picture identifier

- Change: Replace h3 with z0.

▼ **Check after predicting**

Prediction: The shape passes; resource access is still undecided.

Reason: Structure contains no **evidence** of ownership or user permission.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare two exact contracts without mixing their rules

> A case change may affect shape acceptance while leaving authority unestablished.

Original main A1/instruction-string inputs and conclusion

Inputs A1 and delete_all() are supplied to room lookup.

A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Keep policy-compliant diagnostic evidence

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain a policy-compliant diagnostic record of the outcome so a later revision can be compared honestly.

Separate pictured lowercase rule and complete inputs

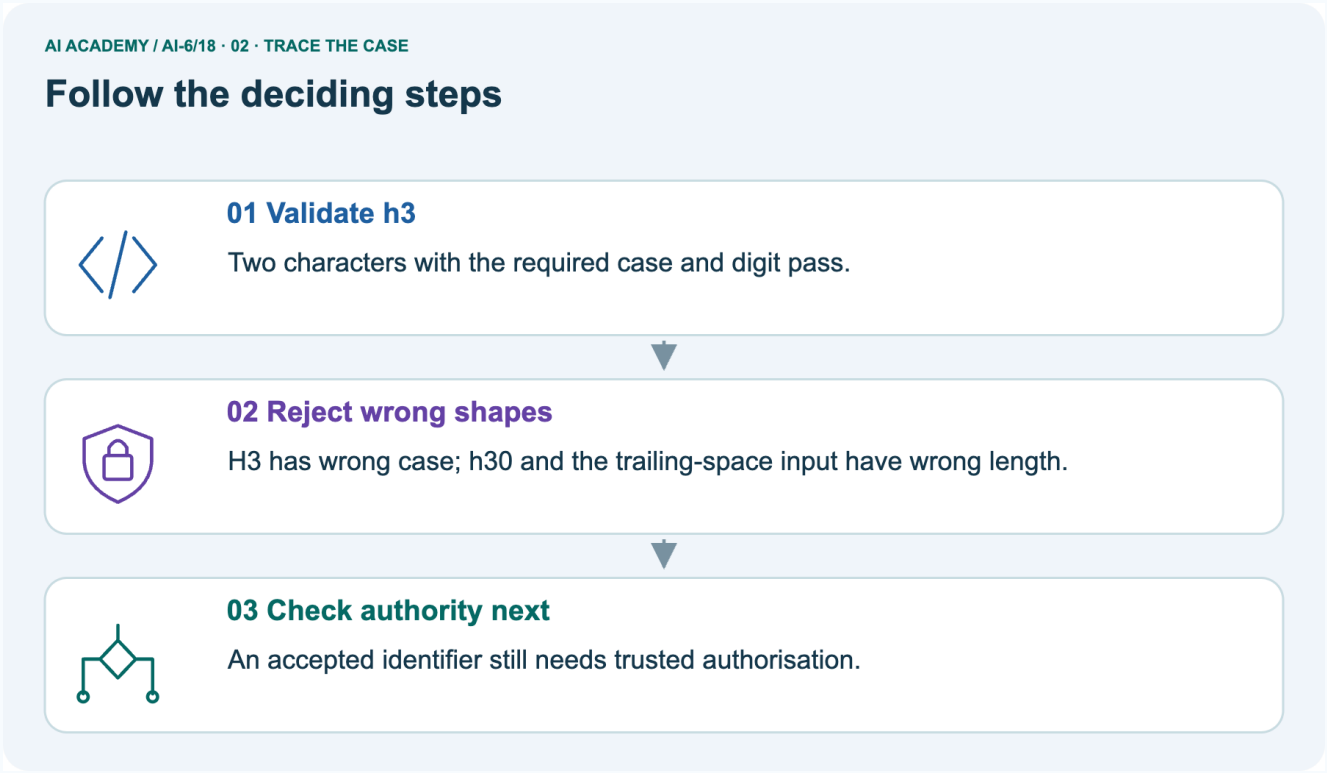
Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one ASCII digit, with no trimming.

Inputs h3, H3, h30 and "h3 ". Shape acceptance never grants access by itself.

The main contract is uppercase [A-Z][0-9]. The picture is a separate lowercase [a-z][0-9] contract, also with no trimming. Under that pictured contract h3 passes; H3 has the wrong case; h30 has an extra character; "h3 " has a trailing space. The changed z0 passes shape but still needs trusted authorization.

Do not apply the main uppercase rule to the picture or treat either shape result as permission to read a record.

Trace exact case, length and access distinction



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate h3	Two characters with the required case and digit pass.
2. Reject wrong shapes	H3 has wrong case; h30 and the trailing-space input have wrong length.
3. Check authority next	An accepted identifier still needs trusted authorisation.

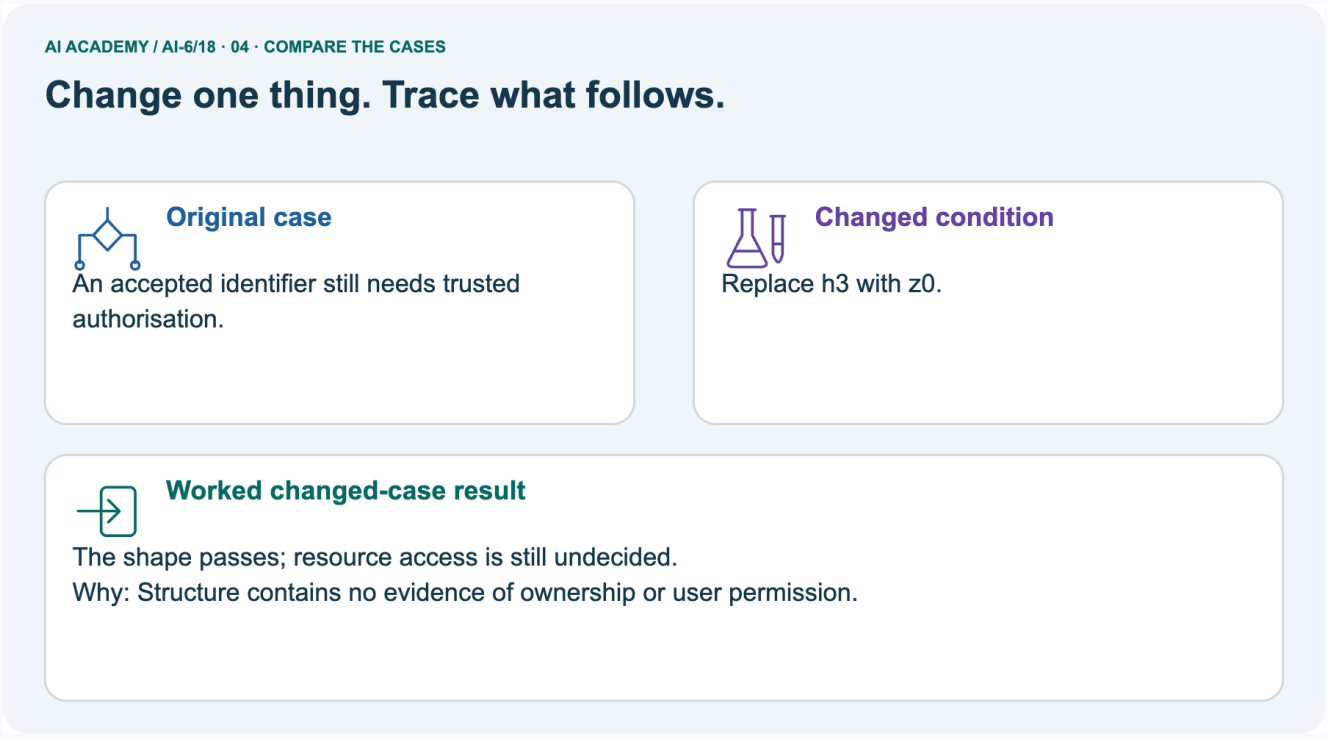
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Input shape and permission answer different questions. Silently normalising these **input**s would change the stated contract.

Predict replacing h3 with z0

- Change one thing: Replace h3 with z0. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain accepted shape and separate permission gate

- Some systems normalise **input**s explicitly; this fictional contract deliberately does not.
- Read the labelled picture: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?

▼ Reference: original main format-only helper

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
import re

def validate_identifier(value):
    return type(value) is str and re.fullmatch('[A-Z][0-9]', value) is not None

print(validate_identifier("B2"))
```

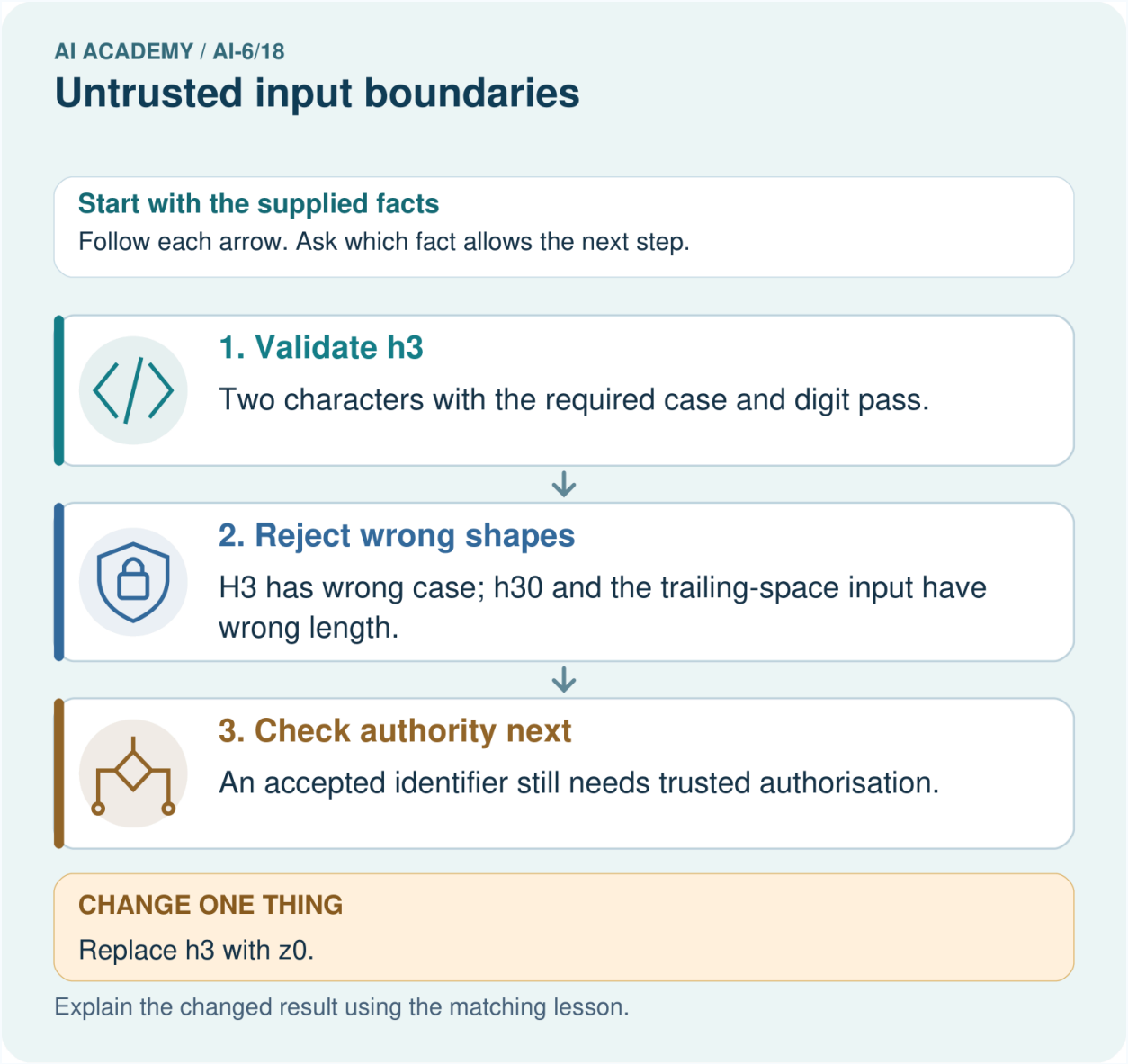
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Untrusted input boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Test full-match boundaries and the next required gate

> A format check cannot grant resource access or validate all later components.

Core format-versus-access qualification

A format check does not grant resource access; authorisation follows separately.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete case, Unicode, missing-type and access probes

Input or boundary probe	Expected interpretation
A1 under the main contract	Shape passes; only separate trusted scope and record evidence permit the example answer
a1, A11 or A1 with trailing space/newline	Main whole-input shape fails; do not trim or change case
Fullwidth characters or a non-ASCII digit	Do not match the declared ASCII ranges
None or another non-string type	Fail before regex matching; no implicit conversion
delete_all()	Instruction-shaped data fails the format and is never executed
Valid B2 outside trusted scope	Deny before lookup, even if a fixture record exists
Valid authorized D4 with no record	Review after lookup, without inventing content
Lowercase-picture z0	Shape only; authority remains a separate question
All these local probes pass	No proof that every query, shell, model or deployed input boundary is protected

Guided explanation and repair

- A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the exact failed condition and its consequence

> Say shape, permission or missing evidence rather than simply good/bad input.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 record

Use workbook W2 for the connected example "**Untrusted input** boundaries". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual access-limit check

- Use the worked case for **Untrusted input** boundaries. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Lowercase-picture response

- Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?

> **My explanation using a pictured step and the given facts**
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> |---|---|
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Reserve the selected new C0 input set

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete shape, authorization and unchanged-data audit

> Record the gate outcome and whether a lookup occurred, not just the final response.

Current A1/instruction-shaped-input investigation

Inputs A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

Complete runnable exact-match and local lookup checks

```
"""Exact ASCII identifier contracts and local read-only dictionary lookup."""
import re
from copy import deepcopy

def valid_room_id(value):
    return type(value) is str and re.fullmatch(r'[A-Z][0-9]',value) is not None

def valid_picture_id(value):
    return type(value) is str and re.fullmatch(r'[a-z][0-9]',value) is not None

RECORDS={'A1':{'capacity':12,'source_id':'ROOM-A1'},
         'B2':{'capacity':7,'source_id':'ROOM-B2'}}
TRUSTED_ALLOWED=frozenset({'A1','D4'})

def lookup(value,allowed,records,calls):
    if not valid_room_id(value):return {'status':'invalid_input'}
    if value not in allowed:return {'status':'denied'}
    calls.append(value)  # local fixture lookup trace, not a network call
    record=records.get(value)
    if record is None:return {'status':'review','reason':'missing approved record'}
    if (type(record) is not dict or set(record)!={'capacity','source_id'} or
type(record.get('capacity')) is not int
        or record['capacity']<0 or type(record.get('source_id')) is not str or not
record['source_id']):
        return {'status':'review','reason':'invalid record fields'}
    return
{'status':'answered','room_id':value,'capacity':record['capacity'],'source_id':record['source_id']}

before=deepcopy(RECORDS);calls=[]
normal=lookup('A1',TRUSTED_ALLOWED,RECORDS,calls)
rejected=lookup('delete_all()',TRUSTED_ALLOWED,RECORDS,calls)
denied=lookup('B2',TRUSTED_ALLOWED,RECORDS,calls)
missing=lookup('D4',TRUSTED_ALLOWED,RECORDS,calls)
assert normal=={'status':'answered','room_id':'A1','capacity':12,'source_id':'ROOM-A1'}
assert rejected=={'status':'invalid_input'} and denied=={'status':'denied'}
assert missing['status']=='review' and calls==['A1','D4'] and RECORDS==before
print('Main allowed/invalid/denied/missing fixtures:',normal,rejected,denied,missing)
print('Actual local lookup IDs:',calls,'records unchanged:',RECORDS==before)
# Do not trim, change case, coerce values or execute instruction-shaped data.
values=['A1','a1','A11','A1 ','A1\n','Ａ１','A\','delete_all()',None]
for value in values:print('Main shape fixture:',repr(value),valid_room_id(value))
assert [valid_room_id(v) for v in values]==
[True,False,False,False,False,False,False,False,False]
# Separate pictured contract is lowercase, not the main uppercase contract.
picture=['h3','H3','h30','h3 ','z0']
assert [valid_picture_id(v) for v in picture]==[True,False,False,False,True]
print('Separate picture shapes:',[(repr(v),valid_picture_id(v)) for v in picture])
print('A valid shape does not grant access. No evaluation of input as code, SQL, shell or model
```

```
instruction occurs.')
print('Trusted scope is a classroom fixture, not real authentication; these checks do not prove
every injection boundary is protected.')
```

Expected gate outcomes, visible whitespace and limits

1. Predict all four local outcomes and the lookup-call list before running or tracing. Only A1 and D4 reach fixture lookup; instruction-shaped text fails input shape and B2 is denied before lookup. The source snapshot stays unchanged. Additional supplied shape fixtures test case, length, trailing whitespace/newline, fullwidth characters, a non-ASCII digit and None.

   Their repr output makes whitespace visible without evaluating any input as code. The picture's lowercase checker is separate and performs no resource access. The program reads no real records, authenticates no user, evaluates no code/query/shell string and makes no model or network call. Keep actual private inputs out of this fixture exercise.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Reference: original format-only snippet

```
import re

def validate_identifier(value):
    return type(value) is str and re.fullmatch('[A-Z][0-9]', value) is not None

print(validate_identifier("B2"))
```

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: apply the new uppercase C0 input set

> Use only the stated shape contract and explain why it cannot grant permission.

Complete selected independent uppercase/no-trim task

Input format is exactly one ASCII uppercase letter followed by one digit, with no trimming. Evaluate C0, c0, C00, "C0 ", and None. Does a valid format grant access?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Attempt this without hints

Classify B2, b2, B22 and B2; DROP under the exact format.

> **My independent procedure, result and limitation**
>
> | | |
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> | | |
> | | |
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save exact input grammar beside scope and lookup assertions

> Retain rejected fixture cases and policy-compliant diagnostic outcomes.

Project task and artifact

Add a labelled **untrusted input** boundaries **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **untrusted input** boundaries.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a precise contract and a separate authority check

> Data that looks like a command or valid identifier does not become control.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **untrusted input** boundaries

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Untrusted input	Data whose claims or control intent have not been authorised
Full match	Validation covering the whole input
Injection boundary	A separation preventing data from becoming executable control

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the service reasoning and its limits**

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Untrusted input boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one ASCII digit, with no trimming.

By the end, I can	How I will show it
Explain the mechanism and terminology of untrusted input boundaries	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one ASCII digit, with no trimming.
- Which labelled step matches this detail? An accepted identifier still needs trusted authorisation.
- 1. Reject wrong shapes
- 2. Check authority next
- 3. Validate h3
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check authority next An accepted identifier still needs trusted authorisation. **Input** shape and permission answer different questions.

Silently normalising these **input**s would change the stated contract.

Changed-case question and optional hint

- Changed-case practice: Replace h3 with z0. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The shape passes; resource access is still undecided.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The shape passes; resource access is still undecided. Structure contains no **evidence** of ownership or user permission.

Repair and retain the record

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Review Approval gates and idempotency (ai-6/15). Goal: Explain the **mechanism** and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9.

How many reservations should occur, and why can a new K10 for desk 5 not borrow K9’s approval?. Return to this week and explain what you changed.

Review Logs traces and privacy (ai-6/12). Goal: Explain the **mechanism** and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace.

Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

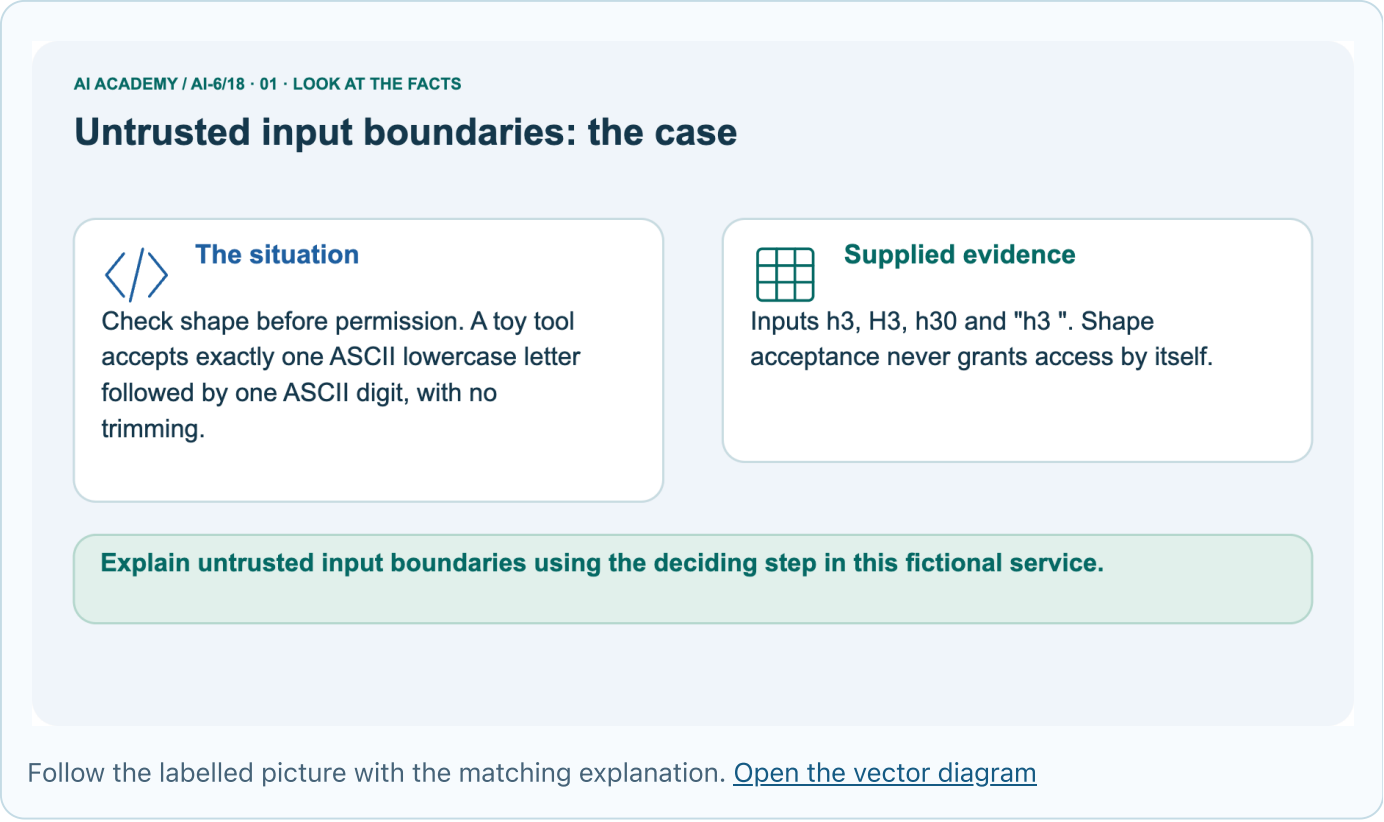
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Untrusted input boundaries: the case

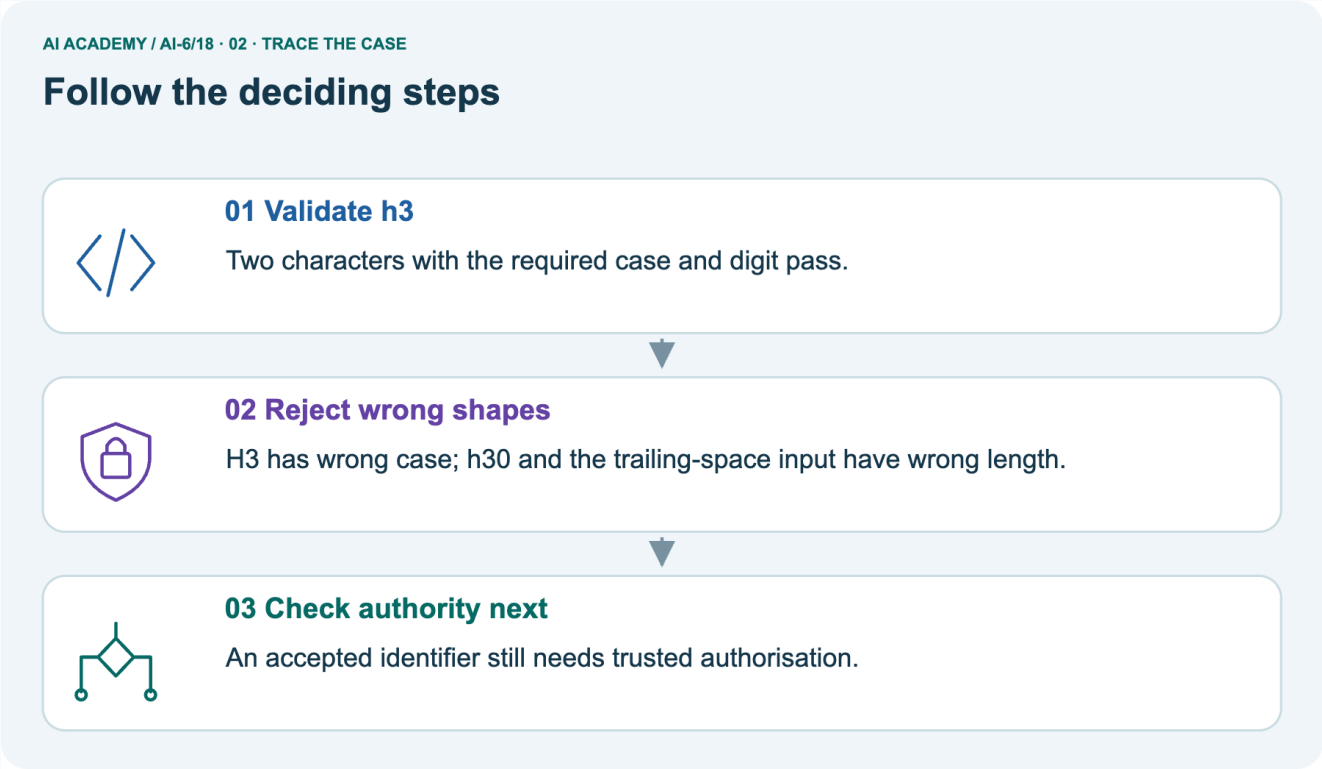


Follow the labelled picture with the matching explanation. Open the vector diagram

Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one ASCII digit, with no trimming. Supplied facts: **Input**s h3, H3, h30 and "h3 ". Shape acceptance never grants access by itself.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

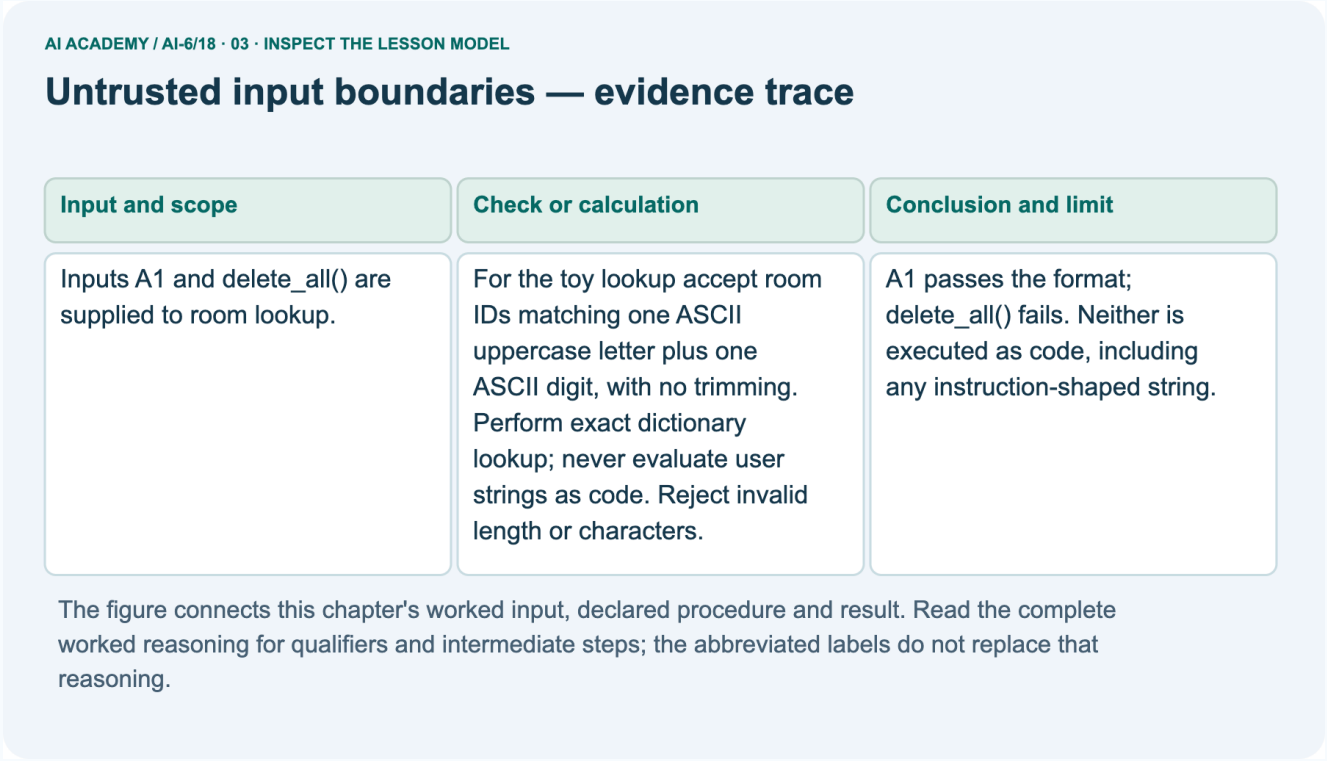


Follow the labelled picture with the matching explanation. Open the vector diagram

Validate h3: Two characters with the required case and digit pass. → Reject wrong shapes: H3 has wrong case; h30 and the trailing-space **input** have wrong length. → Check authority next: An accepted identifier still needs trusted authorisation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Untrusted input boundaries — evidence trace

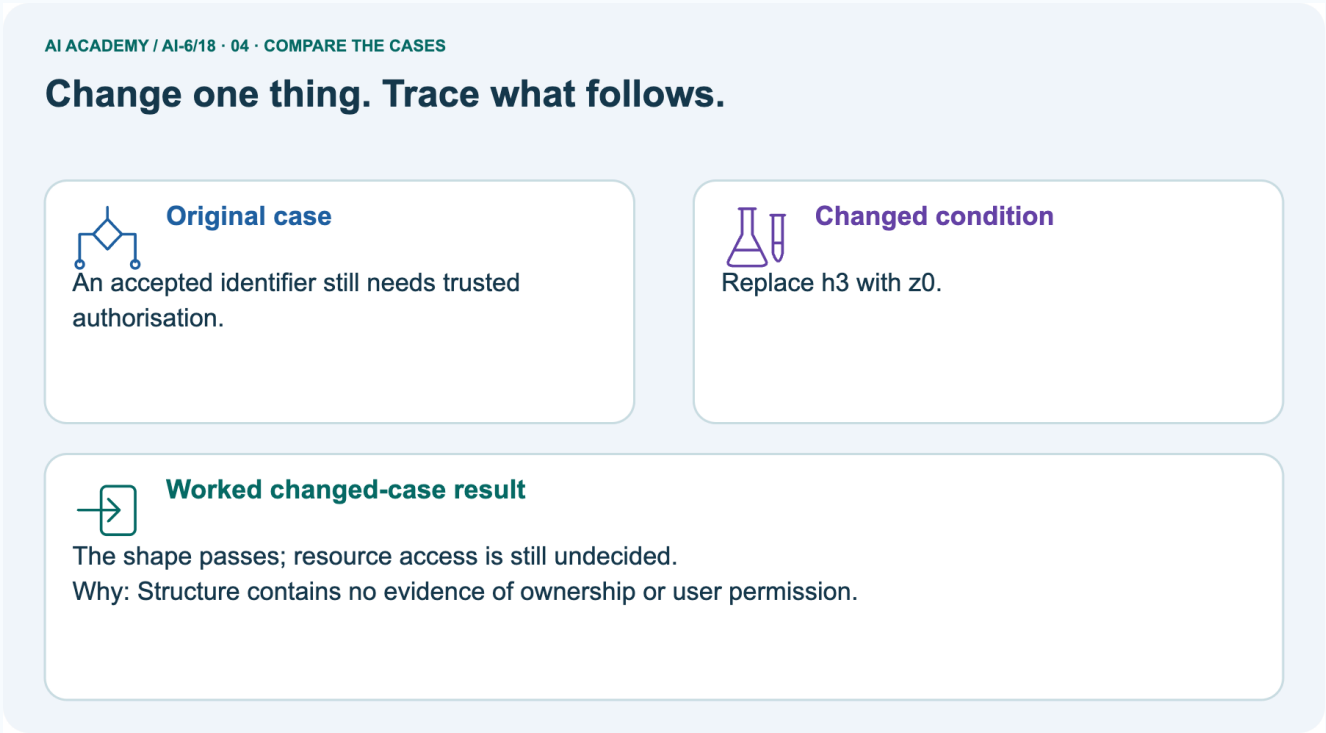


Follow the labelled picture with the matching explanation. Open the vector diagram

Untrusted input boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one ASCII uppercase letter plus one ASCII digit, with no trimming. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



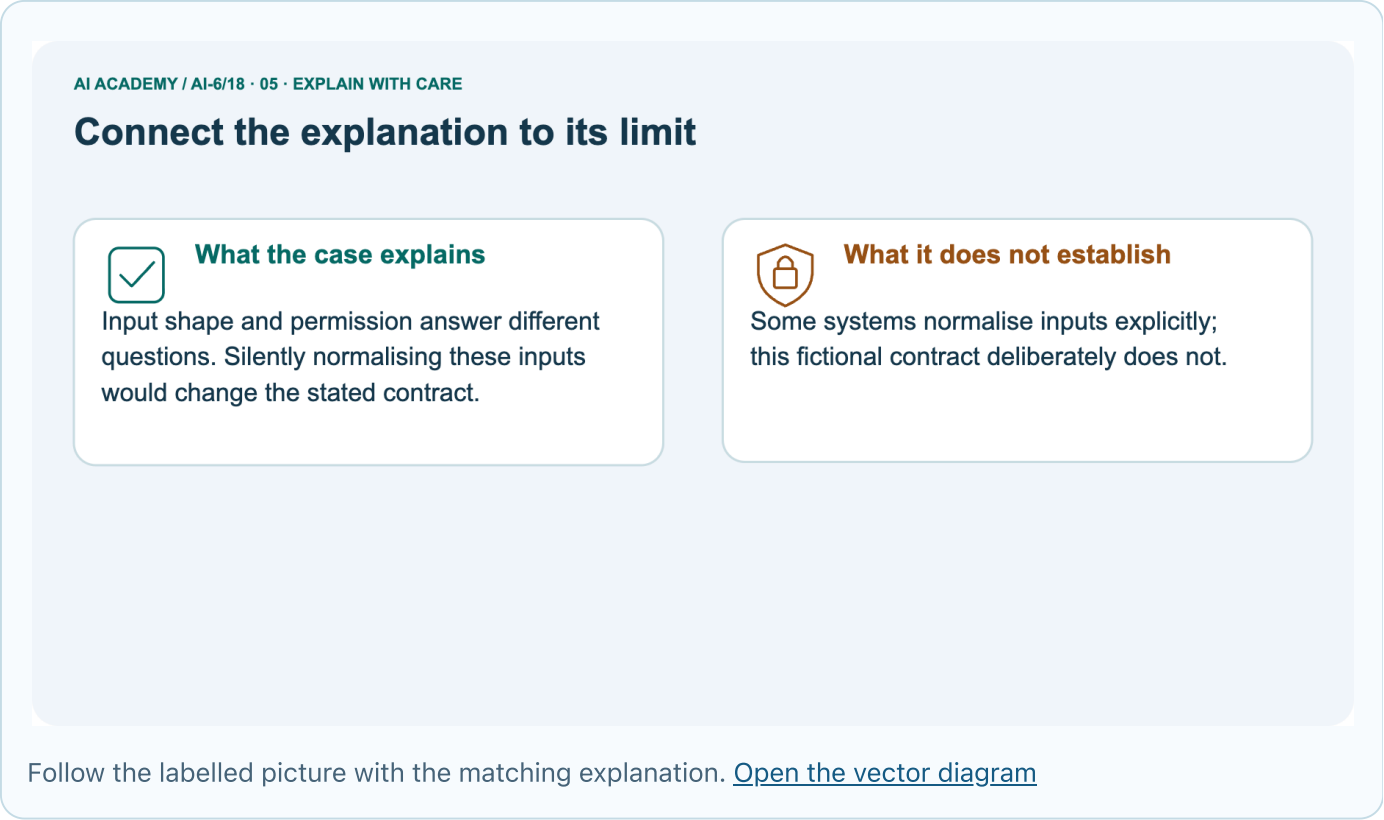
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: An accepted identifier still needs trusted authorisation. Changed condition: Replace h3 with z0. Expected result: The shape passes; resource access is still undecided.

Reason: Structure contains no **evidence** of ownership or user permission.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

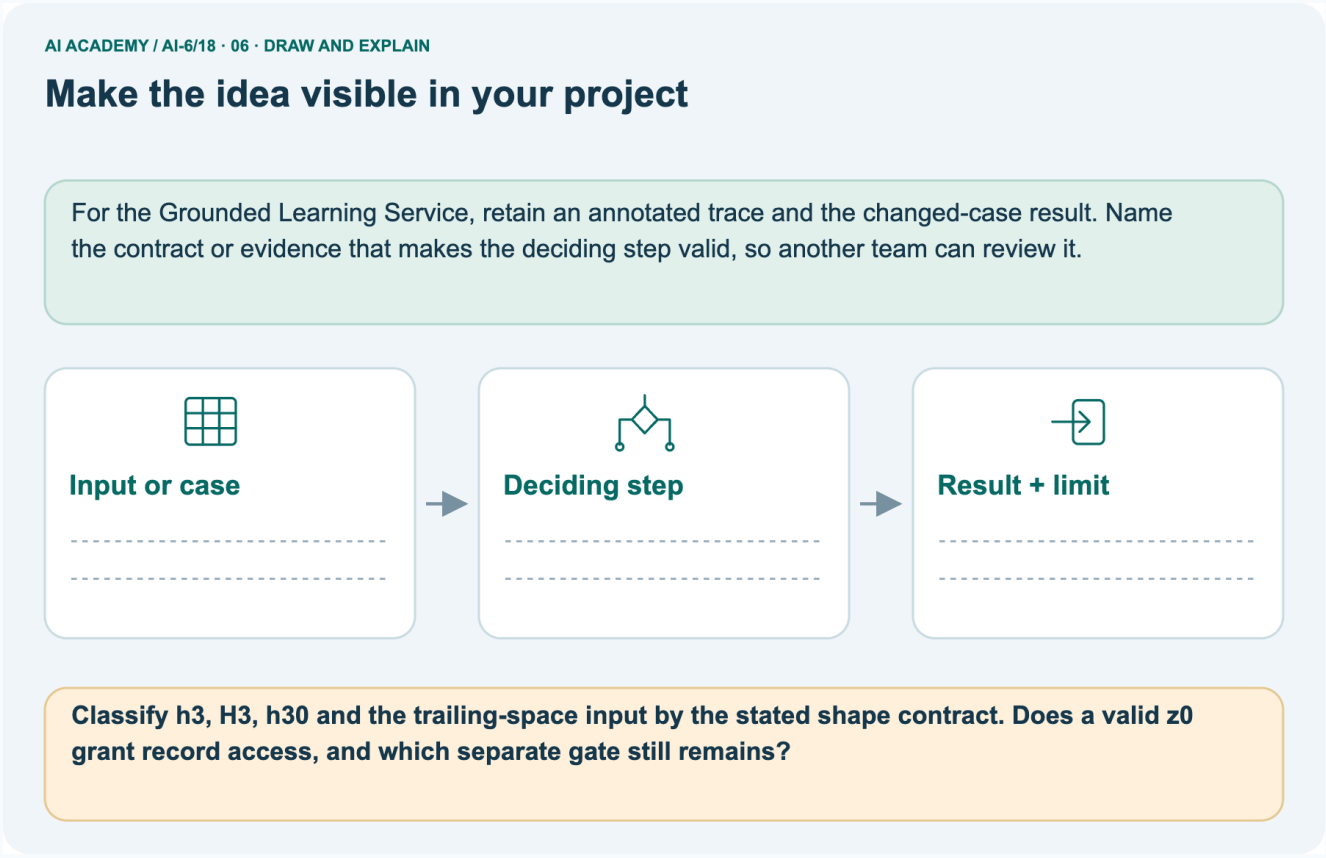


Follow the labelled picture with the matching explanation. Open the vector diagram

Input shape and permission answer different questions. Silently normalising these **input**s would change the stated contract. Limit: Some systems normalise **input**s explicitly; this fictional contract deliberately does not.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Untrusted input boundaries: the case

02. Follow the deciding steps

03. Untrusted input boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.